Chapter 8: Blood-Borne Diseases

Intended Learning Objectives

By the end of this chapter, students should be able to:

1. **Define blood-borne diseases and explain their transmission routes.**
2. **Describe the epidemiology, clinical progression, and complications of HBV, HCV, and HIV.**
3. **Identify high-risk populations and occupational exposure risks.**
4. **Apply prevention strategies including vaccination, screening, and post-exposure prophylaxis.**
5. **Interpret surveillance data and contribute to public health control programs.**

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1. Introduction

Blood-borne diseases are infections transmitted through exposure to infected blood or body fluids. They represent a major public health concern due to their potential for chronic illness, cancer, and death. Healthcare workers, intravenous drug users, and recipients of unscreened blood products are particularly vulnerable. The three most significant blood-borne pathogens are:

- **Hepatitis B virus (HBV)**
- **Hepatitis C virus (HCV)**

- **Human Immunodeficiency Virus (HIV)**

These diseases share common transmission routes but differ in their natural history, treatment options, and prevention strategies.

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2. Modes of Transmission

Exposure Route	Examples
Percutaneous	Needle-stick injuries, cuts with contaminated sharps
Mucosal	Splash to eyes, nose, or mouth with infected fluids
Sexual contact	Unprotected intercourse, especially with multiple partners
Vertical transmission	From mother to child during birth or breastfeeding
Blood transfusion	Especially in settings with poor screening protocols
Shared injection equipment	Among intravenous drug users
Unsafe medical/dental practices	Reuse of needles, poor sterilization

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3. Hepatitis B Virus (HBV)

3.1 Causative Agent

HBV is a double-stranded DNA virus from the Hepadnaviridae family. It is highly infectious—100 times more than HIV—and can survive outside the body for up to 7 days.

3.2 Epidemiology

- **Global burden: ~296 million chronic carriers (WHO, 2023)**
- **Endemic in Africa, Asia, and parts of the Middle East**
- **Egypt: Intermediate endemicity; universal infant vaccination introduced in 1992**

3.3 Incubation Period

30–180 days (average 60–90 days)

3.4 Clinical Features

- **Acute phase: Fever, malaise, nausea, jaundice**
- **Chronic phase: Often asymptomatic; may progress to cirrhosis or hepatocellular carcinoma**
- **Fulminant hepatitis: Rare but life-threatening**

3.5 Diagnosis

- **HBsAg: Indicates current infection**
- **Anti-HBs: Indicates immunity (natural or vaccine-induced)**
- **HBeAg and HBV DNA: Assess infectivity and viral replication**
- **Liver function tests: ALT, AST elevation**

3.6 Prevention

- **Vaccination: 3-dose schedule; highly effective**
- **Post-exposure prophylaxis: HBIG + vaccine within 24 hours**
- **Safe practices: Blood screening, sterile equipment, education**

3.7 Public Health Notes

- **HBV vaccine is part of Egypt's Expanded Program on Immunization (EPI)**
- **Screening of pregnant women prevents vertical transmission**
- **Chronic carriers require regular monitoring and antiviral therapy**

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4. Hepatitis C Virus (HCV)

4.1 Causative Agent

HCV is a single-stranded RNA virus from the Flaviviridae family. It has multiple genotypes; genotype 4 is most common in Egypt.

4.2 Epidemiology

- **Global burden: ~58 million chronic cases (WHO, 2023)**
- **Egypt had one of the highest prevalence rates globally due to past schistosomiasis campaigns**
- **National screening and treatment programs have significantly reduced incidence**

4.3 Incubation Period

2 weeks to 6 months

4.4 Clinical Features

- **Acute phase: Often asymptomatic or mild flu-like symptoms**
- **Chronic phase: Fatigue, liver enzyme elevation**

- **Complications: Cirrhosis, hepatocellular carcinoma**

4.5 Diagnosis

- **Anti-HCV antibodies: Initial screening**
- **HCV RNA (PCR): Confirms active infection**
- **Genotyping: Guides treatment**
- **Fibroscan or biopsy: Assesses liver damage**

4.6 Prevention

- **No vaccine available**
- **Blood safety: Rigorous screening of donors**
- **Harm reduction: Needle exchange programs, education**
- **Infection control: Sterile medical procedures**

4.7 Public Health Notes

- **Egypt's 100 Million Health Initiative screened millions for HCV**
- **Direct-acting antivirals (DAAs) offer >95% cure rate**
- **WHO aims to eliminate HCV as a public health threat by 2030**

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5. Human Immunodeficiency Virus (HIV/AIDS)

5.1 Causative Agent

HIV is a retrovirus that targets CD4+ T lymphocytes, leading to progressive immune deficiency.

5.2 Epidemiology

- **Global burden: ~39 million people living with HIV (UNAIDS, 2023)**
- **Egypt: Low prevalence (<0.1%), but increasing among key populations**
- **High-risk groups: MSM, sex workers, IV drug users**

5.3 Incubation Period

Weeks to years; AIDS may develop 5–10 years after infection

5.4 Clinical Features

- **Acute phase: Fever, rash, lymphadenopathy**
- **Chronic phase: Asymptomatic or mild symptoms**
- **AIDS: Opportunistic infections (TB, PCP), malignancies (Kaposi's sarcoma), wasting**

5.5 Diagnosis

- **Rapid test or ELISA: Screening**
- **Western blot or PCR: Confirmation**
- **CD4 count and viral load: Monitor disease progression and treatment response**

5.6 Prevention

- **No vaccine available**
- **Safe sex practices: Condom use, education**
- **PEP: Antiretroviral therapy within 2 hours of exposure**
- **PrEP: Daily antiretrovirals for high-risk individuals**

- **Blood safety and sterile equipment**

5.7 Public Health Notes

- **UNAIDS 95-95-95 targets: 95% diagnosed, 95% on treatment, 95% virally suppressed**
- **Stigma remains a barrier to care**
- **Integration with TB and reproductive health services improves outcomes**

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6. Summary Table

Disease	Agent	Incubation	Chronic Risk	Vaccine	Key Prevention	Treatment
HBV	DNA virus	30–180 days	Yes	Yes	Vaccine, HBIG	Antivirals
HCV	RNA virus	2–24 weeks	Yes	No	Blood safety	DAAs
HIV	RNA virus	Variable	Yes	No	Safe sex, PEP	ART

✅ True/False Questions

1. **Hepatitis B is more infectious than HIV and can survive outside the body for several days.**
   True
2. **There is an effective vaccine available for Hepatitis C.**
   False

3. **HIV infection always presents with symptoms in the early phase.**
   False
4. **Post-exposure prophylaxis for HIV should be started within 72 hours of exposure.**
   True
5. **HCV infection can be cured with direct-acting antiviral therapy.**
   True